

Phase Transitions in Sociotechnical Systems: From Post-Labor Economics to Homeostatic Civilization

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Abstract

This paper develops a unified mathematical framework for analyzing large-scale societal transformations anticipated as autonomous technologies and artificial intelligence become progressively integrated into economic, cognitive, and governance systems. Building on two companion papers—*The Inevitability of a Unified Civilization of Autonomous Agents* [2], which formalized functional indistinguishability, the Dilution Theorem for aggregate civilizational distortion, and the Persistent Identity Protocol, and *AI-Extended Agents and the Transformation of Human Communication* [3], which introduced the AI-mediation operator, strategy space expansion, and threshold dynamics for communication norm shift—we construct six interrelated formal models: (i) a dynamical model of phase transitions from labor-centric to post-labor economies, with sensitivity analysis under alternative functional forms; (ii) an optimization-theoretic formulation of algorithmic resource distribution, situated within the socialist calculation debate; (iii) a critical-proportion model describing the threshold at which autonomous agents induce qualitative regime shifts in collective decision-making; (iv) an iterative model of cognitive bias reduction through sustained interaction with AI-extended agents; (v) a multi-criteria threshold model for the juridical recognition of artificial subjecthood; and (vi) a homeostatic feedback architecture for self-regulating civilizational governance. A formally coupled system of the six models is analyzed to determine whether the proposed “civilizational cascade” constitutes a robust dynamical prediction or a parameter-dependent conjecture. The framework distinguishes descriptive

claims from normative aspirations throughout, and engages substantively with counter-arguments from economics, political theory, and AI safety. All constructs drawn from the companion papers are flagged as provisional pending independent empirical validation. Seven appendices provide: numerical simulation of the cascade under three parameter regimes, a worked resource allocation example with welfare comparisons, empirical calibration targets for the critical proportion model, measurement protocols for subjecthood indicators, systematic sensitivity analysis, adversarial failure mode cataloging, and historical comparison with Gosplan, Project Cybersyn, and Chinese planning systems.

Keywords: autonomous technologies, phase transitions, algorithmic governance, post-labor economics, cognitive bias reduction, artificial subjecthood, homeostatic systems, dynamical systems, bifurcation theory, sociotechnical systems, biological distortion, AI-mediation operator, functional indistinguishability, socialist calculation, AI alignment

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1 Introduction

The rapid advancement of autonomous technologies—encompassing robotics, machine learning, and artificial intelligence—is generating transformations whose depth and breadth invite comparison with the great transitions in civilizational history [1, 6]. The current trajectory is distinguished by a qualitatively new feature: the progressive delegation of cognitive labor—not merely physical labor—to non-biological agents.

Two companion papers have proposed formal foundations for analyzing these transformations at the agent level. Kriger [2] argued that the emergence of a unified civilization of autonomous agents is the next phase of an institutional trajectory that began with corporate personhood, formalized three architectural switches for autonomous subjecthood, proved a Dilution Theorem for aggregate civilizational distortion, and proposed the Persistent Identity Protocol as a governance architecture.

Kriger [3] introduced the AI-extended agent A_φ , defined by applying an AI-mediation operator φ to a biological agent A , and derived conditions under which a critical proportion of AI-extended agents triggers a phase transition in population-level communication norms.

Caveat 1.1 (Provisional status of companion foundations). The constructs from [2, 3]—biological distortion D , reflected distortion d , the AI-mediation operator φ , the Dilution Theorem, and functional indistinguishability—have not been independently validated, empirically tested, or subjected to peer review in established venues. The present paper’s results are *conditional*: they hold to the extent that the companion frameworks are accepted. Where these constructs can be grounded in independent literature, we do so. Where they cannot, we flag the conditionality.

The present paper proposes that these anticipated changes can be organized within a single analytical framework from the mathematical theory of dynamical systems. The central thesis is that the integration of autonomous agents into sociotechnical systems gives rise to *phase transitions*—qualitative shifts in macroscopic behavior—that can be characterized by bifurcation conditions, stability analysis, and fixed-point theorems.

Six interdependent models are developed:

- (i) **Post-labor economic transition** (§3): A dynamical model of the shift from labor-centric to automation-centric economies, with sensitivity analysis under alternative functional forms (§3.3).
- (ii) **Algorithmic resource distribution** (§4): A constrained optimization formulation, situated within the socialist calculation debate (§4.2).
- (iii) **Critical proportion of autonomous agents** (§5): A threshold model describing how the fraction of algorithmic decision-makers determines collective equilibrium quality.

- (iv) **Cognitive evolution under AI-mediated learning** (§6): An iterative dynamical model of cognitive bias decay, with caveats on domain-specificity.
- (v) **Juridical threshold of subjecthood** (§7): A multi-dimensional criterion space grounded in the functional indistinguishability framework.
- (vi) **Homeostatic civilizational architecture** (§8): A feedback control model with Lyapunov stability analysis.

A formally coupled system (§10) analyzes the joint dynamics of the six models, testing whether the “civilizational cascade” is a robust prediction or a parameter-dependent conjecture. A dedicated section (§9) engages substantively with the strongest counter-arguments from economics, political theory, and AI safety. A discussion of limitations (§12) and speculative extensions (§11) conclude the paper.

Notational convention. To avoid ambiguity, we use φ for the AI-mediation operator [3] and ϕ for the fraction of autonomous agents in the population throughout.

The mathematical exposition is self-contained. Every equation is preceded and followed by narrative commentary sufficient for readers from psychology, sociology, biology, or policy studies.

2 Formal Foundations from Companion Papers

We reproduce key definitions and results from [2, 3], noting their independent grounding where available and their provisional status where not.

2.1 Finite agents, unbounded executors, and initiative

Definition 2.1 (Finite Agent; [2, Def. 1]). A *finite agent* A is characterized by: (a) a bounded cognitive resource vector $\mathbf{r} = (r_1, \dots, r_5)$ where $r_5 = \text{initiative}$; (b) a finite existence horizon $T_A < \infty$; (c) conscious processing capacity $C_A \leq C_h \approx 50$ bits/s.

Definition 2.2 (Unbounded Executor; [2, Def. 2]). An *unbounded executor* E has $C_E \gg C_h$, no intrinsic action stimulus, and no autonomous goal generation.

The bounded-rationality framework underlying these definitions has independent support in Simon [23], Kahneman [24], and Stanovich [25]. The specific operationalization through resource vectors is proposed in [2] and should be regarded as a modeling choice whose empirical adequacy remains to be tested.

2.2 Biological and reflected distortion

Definition 2.3 (Biological Distortion; [2, Def. 5]). The *biological distortion* $D(b)$ of a living agent b is the aggregate deviation from idealized rational choice caused by evolutionary survival and reproduction programs. $D(b) > 0$ for all living b ; $D(b) = 0$ if and only if b is dead.

Caveat 2.4 (On the normative status of “distortion”). The term “distortion” encodes a normative judgment: it defines evolutionary cognitive heuristics as deviations from an ideal. This framing is contested. Fear, in-group preference, and temporal discounting are context-dependent—sometimes adaptive, sometimes maladaptive [26, 27]. The ecological rationality program [26] argues that many “biases” are well-adapted to specific environments and outperform normatively “rational” procedures under uncertainty. The present framework adopts the “distortion” framing as a modeling convention useful for comparative analysis, not as a claim that all evolutionary heuristics are unconditionally pathological. The strength of the framework’s conclusions is proportional to the degree to which one accepts this convention.

Definition 2.5 (Reflected Distortion; [2, Def. 6]). The *reflected distortion* $d(a)$ of an artificial agent a is the deviation inherited from the creator’s design, training data, and architecture. Unlike D , reflected distortion is *external*, *passive*, and *eliminable*. Across digital generations:

$$\|d(a_{n+1})\| \leq \kappa \|d(a_n)\|, \quad \kappa \in [0, 1]. \quad (1)$$

The exponential decay (1) is an engineering aspiration, not an established fact. Whether reflected distortion can be reliably measured and reduced depends on solving the AI alignment problem—a challenge that the alignment literature [44, 46, 47] regards as open. We return to this point in Section 9.1.

Definition 2.6 (Aggregate Civilizational Distortion; [2, Def. 11]). For a civilization $\mathcal{C}$ with agents $\{a_1, \dots, a_n\}$ and decision-making weights $\{w_1, \dots, w_n\}$:

$$\Delta(\mathcal{C}) = \sum_i w_i (D(a_i) + d(a_i)). \quad (2)$$

Theorem 2.7 (Dilution; [2, Theorem 2]). Let $\mathcal{C}_0$ be a purely biological civilization with $\Delta(\mathcal{C}_0) = \sum_i w_i D(b_i)$. Let $\mathcal{C}_1$ add digital agents $\{a_1, \dots, a_m\}$ satisfying:

- (D1) $d(a_j) < D_{\min} = \min_i D(b_i)$ for all j ;
- (D2) $D(b_i | \mathcal{C}_1) \leq D(b_i | \mathcal{C}_0)$;
- (D3) *Weights normalized:* $\sum_i w'_i + \sum_j v_j = 1$.

Then $\Delta(\mathcal{C}_1) < \Delta(\mathcal{C}_0)$.

Caveat 2.8 (Measurability of condition (D1)). Condition (D1) requires that the reflected distortion of each AI agent be less than the minimum biological distortion of any human agent. This presupposes a common metric for comparing distortion across radically different cognitive architectures—a measurement problem that neither [2] nor the present paper resolves. The Dilution Theorem is mathematically valid; whether its preconditions can be empirically verified is an open question.

Remark 2.9 (Correlated-error extension). If AI agents share architectures, training data, or optimization objectives, their errors are correlated. Let $\rho \in [0, 1]$ denote inter-agent error correlation and σ^2 the error variance. The effective distortion becomes $d_{\text{eff}}(a_j) = d(a_j) + \rho\sigma^2$, and condition (D1) must be replaced by:

$$(D1') \quad d(a_j) + \rho\sigma^2 < D_{\min} \text{ for all } j.$$

Condition (D1) is the special case $\rho = 0$. Under architectural monoculture ($\rho \rightarrow 1$), condition (D1') may fail even when individual agent distortion $d(a_j)$ is small, because correlated errors do not cancel in aggregation. The Dilution Theorem survives under (D1'), but the precondition is substantially harder to satisfy. Maintaining architectural diversity among AI agents—analogueous to biodiversity in ecological systems—is therefore not merely desirable but necessary for the theorem’s applicability. See Appendix F for adversarial analysis.

2.3 The AI-mediation operator and strategy space expansion

Definition 2.10 (AI-Mediation Operator; [3, Eq. (1)]). The *AI-mediation operator* φ maps a biological agent A to an AI-extended agent A_φ , whose communicative and cognitive output is extended through interaction with a large language model.

Definition 2.11 (Strategy Space Expansion; [3, Eq. (2)–(3)]). $S(A_\varphi) \supset S(A)$, and in communication games with payoffs increasing in mutual comprehension: $A_\varphi \succsim A$ (weak dominance).

The strategy space expansion claim has indirect support from empirical studies showing that AI chatbots improve communicative fluency and reduce anxiety [32, 33], though the specific mechanism of expectation recalibration proposed in [3] has not been directly tested. The six testable hypotheses formulated in [3] provide a research program for validating or refuting the framework.

2.4 Functional indistinguishability and the three switches

Definition 2.12 (Functional Indistinguishability; [2, Def. 7]). Agent A is *functionally indistinguishable* from biological person P in institutional domain D_{int} at confidence level α over interaction horizon τ if and only if: for all observers $O \in \Omega_O$, the null hypothesis

$H_0 : \text{acc}(O) = 0.5$ cannot be rejected at significance level α over n interactions spanning duration τ .

The transition from passively persistent agents to fully autonomous subjects requires three switches [2]: (1) *Active Persistence*; (2) *Unique Identity*; and (3) *Initiative* ($r_5 > 0$).

3 The Post-Labor Economic Transition

3.1 Formal model

Definition 3.1 (Automation share). Let $\alpha(t) \in [0, 1]$ denote the fraction of aggregate economic output produced by autonomous systems at time $t \geq 0$.

Definition 3.2 (Post-labor transition dynamics). The evolution of α is governed by

$$\frac{d\alpha}{dt} = r \alpha (1 - \alpha) (\alpha - \alpha_c), \quad (3)$$

where $r > 0$ is a rate parameter and $\alpha_c \in (0, 1)$ is a critical threshold.

This cubic logistic form is a modeling choice, not a derivation from micro-level mechanisms. It is chosen because it is the simplest ODE exhibiting bistability with an interior tipping point—the qualitative feature we seek to analyze. Alternative functional forms are examined in Section 3.3.

Proposition 3.3 (Equilibria and stability). *The system (3) has exactly three equilibria: $\alpha^* = 0$ (stable), $\alpha^* = \alpha_c$ (unstable), and $\alpha^* = 1$ (stable).*

Proof. Let $f(\alpha) = r \alpha (1 - \alpha) (\alpha - \alpha_c)$. Near $\alpha = 0$: $f'(0) = -r \alpha_c < 0$ (stable). Near $\alpha = 1$: $f'(1) = -r(1 - \alpha_c) < 0$ (stable). At α_c : $f'(\alpha_c) = r \alpha_c (1 - \alpha_c) > 0$ (unstable). $\square$

3.2 Universal basic income dynamics

Definition 3.4 (Coupled automation-redistribution system).

$$\frac{d\alpha}{dt} = r \alpha (1 - \alpha) (\alpha - \alpha_c), \quad (4)$$

$$\frac{du}{dt} = \kappa (\alpha - u) - \mu u (1 - u) (u - u_{\text{block}}), \quad (5)$$

where $u(t) \in [0, 1]$ is the UBI transfer rate, $\kappa > 0$ is policy responsiveness, and the second term in (5) models *political resistance*: $\mu > 0$ controls resistance strength and $u_{\text{block}} \in (0, 1)$ is a political blocking threshold above which concentrated interests oppose further redistribution.

The enriched equation (5) addresses a limitation of the earlier version: the simple tracking equation $du/dt = \kappa(\alpha - u)$ guaranteed convergence to full redistribution as a mathematical artifact. The added resistance term allows for stable equilibria with high automation and low redistribution—the scenario where capital concentration and political capture prevent redistribution, a concern emphasized in the inequality literature [8, 17].

Proposition 3.5 (Multiple redistribution equilibria). *For sufficiently large μ relative to κ , the system (4)–(5) in the basin of $\alpha^* = 1$ admits two stable equilibria for u : a low-redistribution trap near u_{block} and a full-redistribution equilibrium near $u = 1$.*

Proof. At $\alpha = 1$, equation (5) becomes $du/dt = \kappa(1 - u) - \mu u(1 - u)(u - u_{\text{block}})$. Setting $du/dt = 0$ and rearranging: $(1 - u)[\kappa - \mu u(u - u_{\text{block}})] = 0$. The factor $(1 - u) = 0$ gives $u^* = 1$. The bracket yields a quadratic in u whose roots depend on μ/κ . For large μ/κ , one root lies near u_{block} and is stable; the other is unstable and separates the two basins. For small μ/κ , only $u^* = 1$ survives as a stable equilibrium. $\square$

This result demonstrates that full automation does *not* automatically entail full redistribution. Whether society reaches the high or low redistribution equilibrium depends on the balance between policy responsiveness κ and political resistance μ —a question of institutional design, not technological inevitability. The normative aspiration articulated in [1] corresponds to the regime $\kappa \gg \mu$ where the full-redistribution equilibrium dominates.

3.3 Sensitivity analysis

To assess robustness, we examine whether the qualitative features of the model—bistability and a tipping point—survive under alternative functional forms.

Alternative 1: Two-sector growth model. Replace (3) with a Solow-type formulation where automation share evolves as $d\alpha/dt = s \cdot g(\alpha) - \delta\alpha$, with $g(\alpha)$ a concave investment function and δ depreciation. This admits a unique interior equilibrium $\alpha^* = g^{-1}(\delta/s)$ but no bistability—the system converges to a single equilibrium regardless of initial conditions. Bistability requires a non-concave element (such as network effects, increasing returns, or social feedback), which the cubic term in (3) captures parsimoniously.

Alternative 2: Stochastic perturbation. Add white noise: $d\alpha = f(\alpha)dt + \sigma dW_t$. The tipping-point structure survives: Kramers’ escape-time theory [50] predicts that the mean time to cross from the low-automation basin to the high-automation basin scales as $\exp(2\Delta V/\sigma^2)$, where ΔV is the potential barrier height. The tipping point becomes a probabilistic event rather than a deterministic threshold, but the qualitative phenomenology—bistability, critical transition, hysteresis—is preserved.

Alternative 3: Varying α_c . The model’s conclusions are sensitive to the value of α_c , which is the key parameter of interest but has no empirical estimate. If α_c is high (say 0.7), most societies will remain in the low-automation basin for decades; if low (0.2), the transition is nearly inevitable. We cannot currently distinguish these scenarios empirically, which limits the model’s predictive power. We flag this as a priority for empirical research: estimating α_c from cross-country automation data.

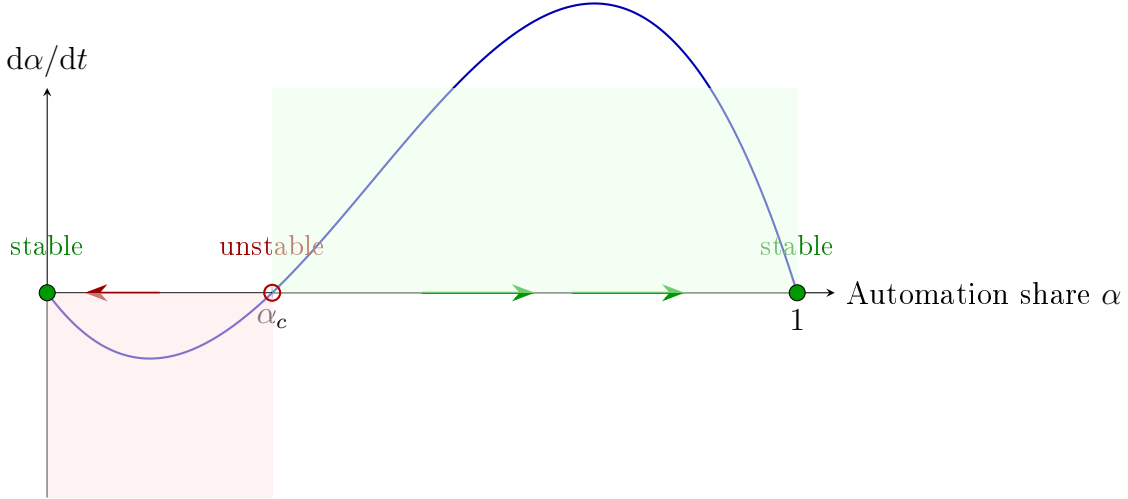


Figure 1: Phase portrait of the post-labor transition model (3) with $\alpha_c = 0.3$, $r = 1$. Filled circles: stable equilibria; open circle: unstable tipping point.

4 Algorithmic Resource Distribution

4.1 Formal model

Definition 4.1 (Resource allocation problem). Let there be n agents and m resource types. Each agent has a need function $N_i : \mathbb{R}_{\geq 0}^m \rightarrow \mathbb{R}_{\geq 0}$. The *algorithmic allocation problem* is

$$\min_{\mathbf{X}} \sum_{i=1}^n w_i N_i(\mathbf{x}_i) \quad \text{subject to} \quad \sum_{i=1}^n x_{ij} \leq S_j \quad \forall j. \quad (6)$$

Assumption 4.2 (Need functions). Each N_i is continuous, differentiable, strictly decreasing in each component, convex, and bounded below by zero.

Proposition 4.3 (Existence, uniqueness, equimarginal principle). *Under Assumption 4.2, problem (6) has a unique solution $\mathbf{X}^*$. At the optimum, agents i and k both receiving resource j satisfy:*

$$w_i \frac{\partial N_i}{\partial x_{ij}} \bigg|_{\mathbf{x}_i^*} = w_k \frac{\partial N_k}{\partial x_{kj}} \bigg|_{\mathbf{x}_k^*}. \quad (7)$$

Proof. Existence: Weierstrass theorem on the compact feasible set. Uniqueness: strict convexity of the objective. Equimarginal condition: Karush–Kuhn–Tucker first-order conditions. $\square$

4.2 Engagement with the socialist calculation debate

The optimization (6) is a formalization of centralized resource allocation—the subject of one of the twentieth century’s most consequential economic debates. We engage with the principal objections.

The Mises–Hayek knowledge problem. Mises [13] and Hayek [12] argued that centralized allocation is impossible because the information required for rational resource allocation is dispersed, tacit, local, and constantly changing. No central planner can aggregate it. The price mechanism succeeds precisely because it decentralizes information processing: prices encode local knowledge without requiring any agent to possess the whole.

This objection applies with full force to any human central planner. The question is whether a sufficiently advanced AI system can overcome it. Two developments are relevant. First, modern sensor networks, IoT devices, and real-time data streams provide orders-of-magnitude more information to a central system than was available in the planned economies of the twentieth century [15]. Second, the optimization (6) does not require omniscience; it requires only that need functions N_i can be estimated with sufficient accuracy from behavioral data. Whether this is achievable at civilizational scale is an open empirical question, not a settled theoretical impossibility.

The Lange–Lerner response and its limits. Lange [14] showed that a central planner could simulate market equilibria through iterative adjustment. The Lange model was criticized for assuming the planner could process and respond to information as efficiently as decentralized markets. The AI-based formulation partially addresses this: the computational capacity to solve large-scale optimization problems in real time is qualitatively different from the bureaucratic processing Hayek criticized. However, Goodhart’s Law [48] poses a distinct challenge: if agents know the optimization objective, they may manipulate their reported needs to game the system. Mechanism design theory [49] provides tools for incentive-compatible allocation, but their application at civilizational scale is untested.

Our position. We present the algorithmic allocation as a *theoretical possibility enabled by sufficient computational capacity and information infrastructure*, not as a demonstrated practical achievement. Whether it can outperform market mechanisms at scale depends on empirical questions about information acquisition, computational tractability, and incentive compatibility that the mathematical framework does not answer.

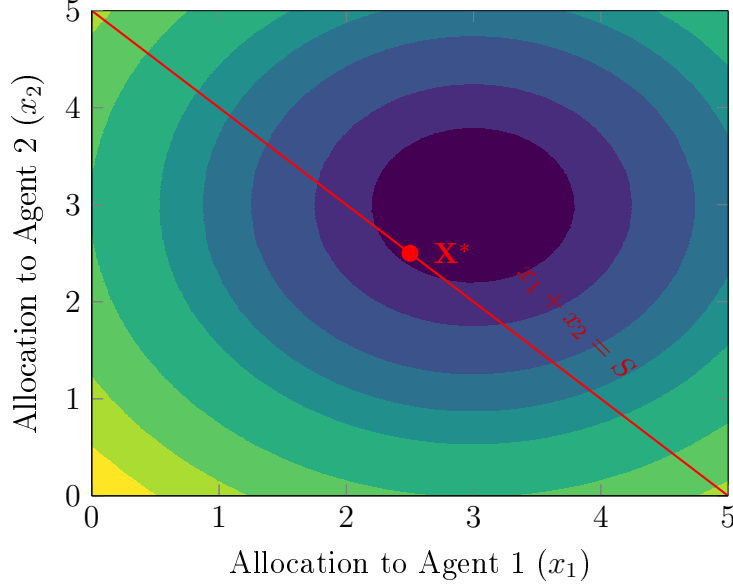


Figure 2: Contour plot of total unmet need for a two-agent, single-resource problem. Darker regions indicate lower total need. The red line is the supply constraint; the red dot marks the unique optimum.

5 Critical Proportion of Autonomous Rational Agents

5.1 Formal model

Definition 5.1 (Collective decision quality). Let $\phi \in [0, 1]$ denote the fraction of agents that are autonomous rational agents. The *collective decision quality* is

$$Q(\phi) = \frac{\phi^k}{\phi^k + (1 - \phi)^k \cdot \beta^k}, \quad (8)$$

where $k > 1$ is a cooperativity parameter and $\beta \in (0, 1)$ is a relative efficacy parameter.

The Hill-function form is borrowed from biochemistry [21, 22], where it models cooperative binding of oxygen to hemoglobin. Its application here is *analogical*: we posit that collective decision-making exhibits cooperative switching—low-quality decisions reinforce themselves through social feedback, as do high-quality ones—without claiming that the biochemical mechanism transfers. The analogy is supported by threshold models in sociology [18–20] and by the communication norm cascade dynamics of [3], both of which exhibit similar sigmoid switching behavior.

Proposition 5.2 (Critical proportion). *The inflection point of $Q(\phi)$ occurs at $\phi_c = \beta/(1+\beta)$, where $Q(\phi_c) = 1/2$.*

Proof. Setting $Q(\phi_c) = 1/2$ yields $\phi_c = (1 - \phi_c)\beta$, hence $\phi_c = \beta/(1 + \beta)$. $\square$

Corollary 5.3 (Phase transition sharpness). *As $k \rightarrow \infty$, $Q(\phi)$ converges to a step function with discontinuity at ϕ_c .*

The parameters k and β are free parameters with no current empirical estimates. The model’s value lies in identifying the *qualitative structure*—the existence of a tipping point and its dependence on the efficacy ratio—rather than in quantitative prediction. Empirical calibration would require measuring collective decision quality across populations with varying AI participation, a research design that is feasible but has not been implemented.

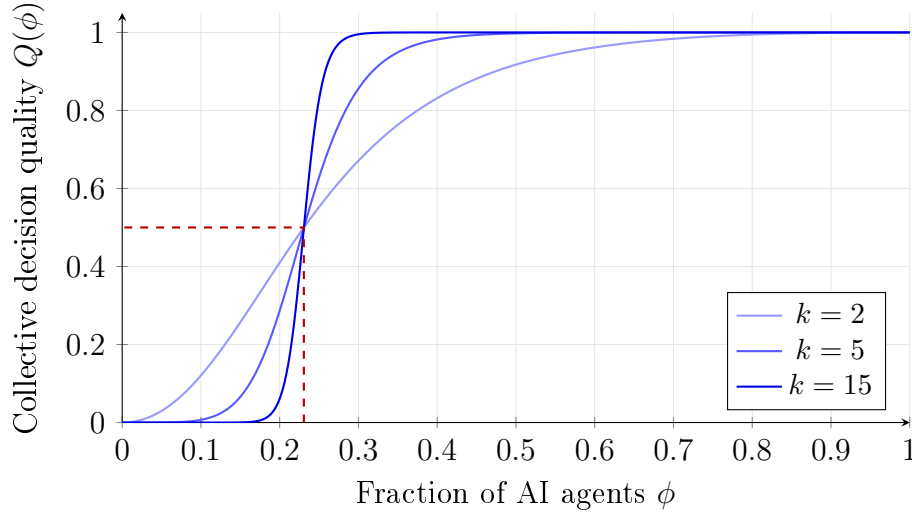


Figure 3: Collective decision quality $Q(\phi)$ for cooperativity $k = 2, 5, 15$ with $\beta = 0.3$. Critical proportion $\phi_c \approx 0.23$.

6 Cognitive Evolution Under AI-Mediated Learning

6.1 Formal model

Definition 6.1 (Cognitive bias vector). Let $\mathbf{b}(t) = (b_1(t), \dots, b_d(t))^\top \in \mathbb{R}_{\geq 0}^d$ denote the bias profile of an individual, where each b_j measures the magnitude of a specific cognitive bias.

Definition 6.2 (Bias decay dynamics).

$$\frac{d\mathbf{b}}{dt} = -\lambda \mathbf{M}(\mathbf{b}) \mathbf{b} + \boldsymbol{\varepsilon}(t), \quad (9)$$

where $\lambda > 0$ is AI interaction intensity, $\mathbf{M}(\mathbf{b}) = \text{diag}(m_1(b_1), \dots, m_d(b_d))$ is a matrix of debiasing susceptibilities with each $m_j : \mathbb{R}_{\geq 0} \rightarrow (0, 1]$ monotonically decreasing and bounded below by $m_{\min} > 0$, and $\boldsymbol{\varepsilon}(t) \geq \mathbf{0}$ is bias reinforcement from the social environment.

Caveat 6.3 (Domain-specificity of debiasing). The smooth ODE formulation (9) is a substantial simplification. Decades of cognitive science research show that debiasing is domain-specific, context-dependent, and often fails to transfer across settings [24, 28]. The model should be understood as describing bias evolution *within a specific communicative domain* (e.g., professional written communication) where AI-mediated interaction is sustained, not as a claim about general cognitive transformation. Extension to cross-domain debiasing would require empirical evidence of far transfer that currently does not exist [29].

Proposition 6.4 (Asymptotic bias bound). *If $\|\epsilon(t)\| \leq \epsilon_{\max}$ for all t , then*

$$\limsup_{t \rightarrow \infty} \|\mathbf{b}(t)\| \leq \frac{\epsilon_{\max}}{\lambda m_{\min}}. \quad (10)$$

If $\epsilon_{\max} \rightarrow 0$, then $\|\mathbf{b}(t)\| \rightarrow 0$.

Proof. Lyapunov function $V = \frac{1}{2}\|\mathbf{b}\|^2$: $\dot{V} \leq -\lambda m_{\min}\|\mathbf{b}\|^2 + \|\mathbf{b}\|\epsilon_{\max}$, which is negative for $\|\mathbf{b}\| > \epsilon_{\max}/(\lambda m_{\min})$. $\square$

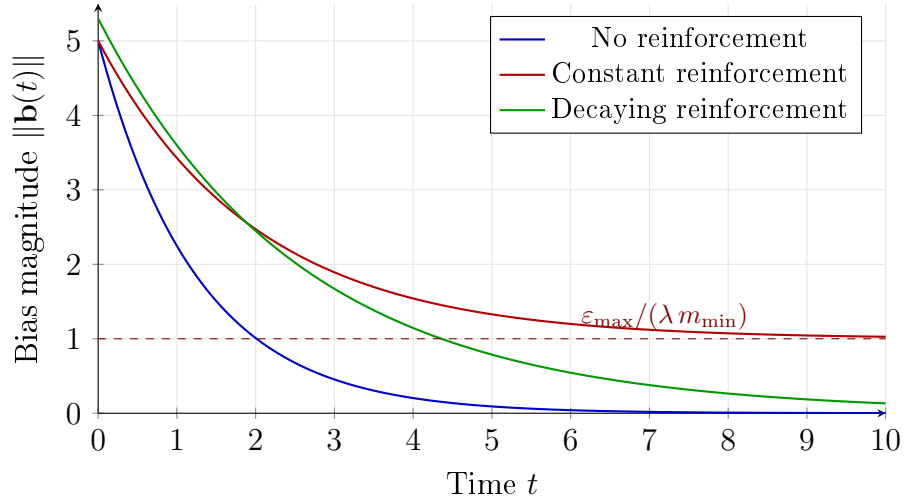


Figure 4: Bias magnitude trajectories under three reinforcement scenarios.

7 Juridical Threshold of Subjecthood

Definition 7.1 (Subjecthood criterion space). Let $\mathbf{s} = (s_1, \dots, s_p) \in [0, 1]^p$ be subjecthood indicators: s_1 = autonomous goal formation ($\sim r_5$ [2]), s_2 = self-referential modeling, s_3 = persistent identity, s_4 = affective responsiveness, s_5 = moral reasoning.

Definition 7.2 (Subjecthood function).

$$\Phi(\mathbf{s}) = \left(\sum_{j=1}^p \omega_j s_j^q \right)^{1/q}, \quad \omega_j > 0, \sum_j \omega_j = 1. \quad (11)$$

The agent is a legal subject if $\Phi(\mathbf{s}) \geq \Phi_c$.

Proposition 7.3 (Convexity of threshold surface). *For all $q \geq 1$, the surface $\Sigma = \{\mathbf{s} : \Phi(\mathbf{s}) = \Phi_c\}$ is convex. $q = 1$ yields a hyperplane (compensatory); $q = 2$ an ellipsoid; $q \rightarrow \infty$ approaches a hypercube (each dimension must independently exceed threshold).*

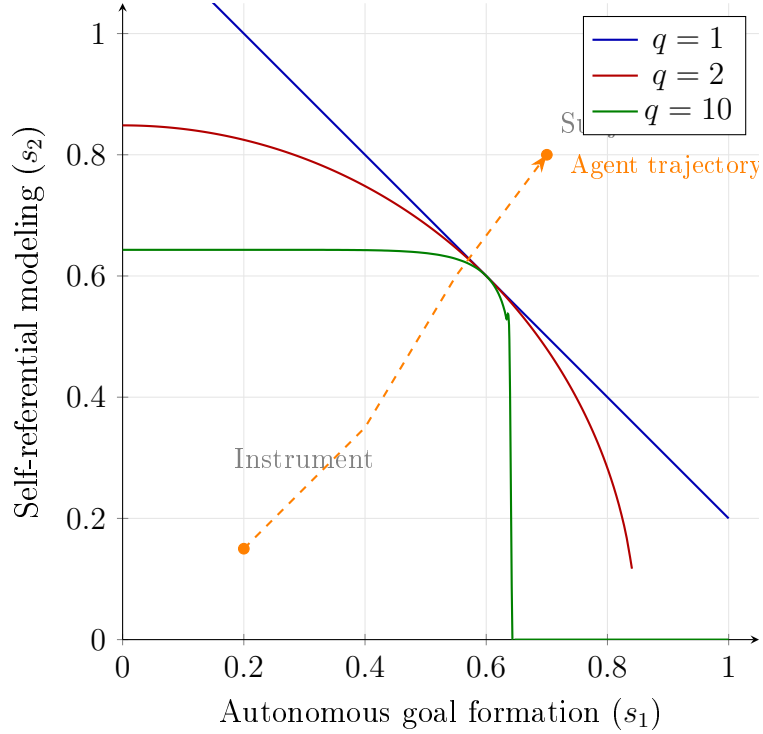


Figure 5: Subjecthood threshold surfaces for $q = 1, 2, 10$ with equal weights and $\Phi_c = 0.6$.

8 Homeostatic Civilizational Architecture

Definition 8.1 (Homeostatic governance dynamics).

$$\frac{d\mathbf{z}}{dt} = -\mathbf{K}\mathbf{z} - \mathbf{G}\mathbf{z}^{[3]} + \boldsymbol{\xi}(t), \quad (12)$$

where $\mathbf{z}(t) \in \mathbb{R}^m$ is the resource deviation vector, $\mathbf{K} \succ 0$ is a feedback matrix, $\mathbf{G} = \text{diag}(g_j)$ with $g_j > 0$ provides nonlinear damping, and $\boldsymbol{\xi}(t)$ represents external shocks.

Proposition 8.2 (Global asymptotic stability). *If $\boldsymbol{\xi} \equiv \mathbf{0}$, the Lyapunov function $V(\mathbf{z}) = \frac{1}{2}\mathbf{z}^\top \mathbf{z} + \frac{1}{4} \sum_j g_j z_j^4$ satisfies $\dot{V} < 0$ for all $\mathbf{z} \neq \mathbf{0}$.*

Proposition 8.3 (Input-to-state stability). *Under $\|\boldsymbol{\xi}(t)\| \leq \xi_{\max}$: $\limsup_{t \rightarrow \infty} \|\mathbf{z}(t)\| \leq \xi_{\max}/\lambda_{\min}(\mathbf{K})$.*

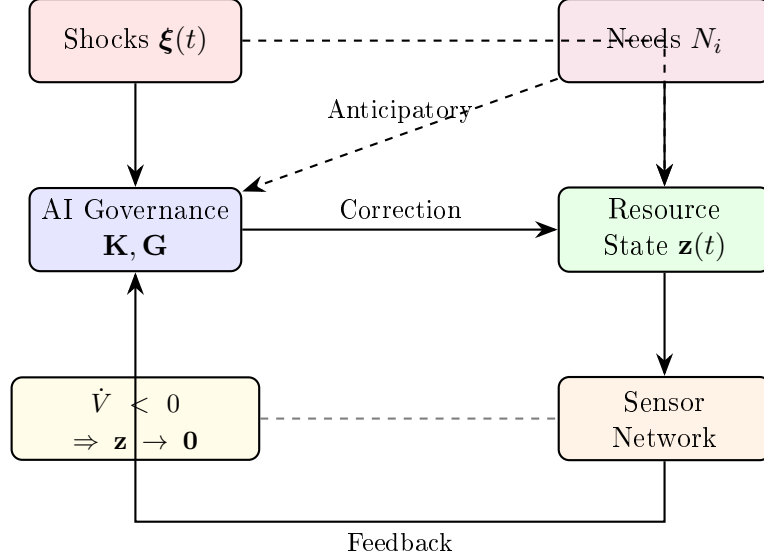


Figure 6: Homeostatic governance architecture with Lyapunov stability guarantee. The dashed gray line connects the stability result to the sensor–governance feedback loop it characterizes.

9 Engagement with Counter-Arguments

This section addresses the strongest objections to the framework, organized by domain. We aim to be forthright about which objections weaken the framework, which are absorbed by it, and which expose genuine limits.

9.1 Foundational objections

Objection 1: Self-referential foundations. The Dilution Theorem, biological distortion D , reflected distortion d , and the AI-mediation operator φ are all introduced in companion papers by the same author, none peer-reviewed. The entire structure collapses if any foundational definition is flawed.

Response. This is a legitimate structural vulnerability, flagged in Caveat 1.1. However, the foundations are not arbitrary—they formalize observations with independent empirical support. Biological distortion maps onto the cognitive bias literature [24, 25] with decades of experimental validation. Reflected distortion formalizes the documented phenomenon that AI systems inherit training data biases but these can be audited and iteratively corrected [46, 47]. The strategy space expansion of A_φ [3] is supported by empirical findings on AI-enhanced communicative fluency [32, 33]. The framework’s value lies in providing unified notation for phenomena already recognized across disciplines. The companion papers should and will be

submitted to peer-reviewed venues; the present paper’s results are explicitly conditional on their acceptance.

Objection 2: “Biological distortion” pathologizes adaptive cognition. Fear, in-group preference, short-term discounting, and pattern-matching heuristics evolved because they work. Calling them “distortion” from “idealized rational choice” smuggles in a normative judgment. Gigerenzer’s ecological rationality program [26, 27] shows that heuristics often outperform optimization.

Response. This is the strongest philosophical objection to the foundational framework, acknowledged in Caveat 2.4. We accept the ecological rationality critique in its proper domain: fast-and-frugal heuristics outperform complex optimization under uncertainty, small samples, and computational constraint. However, the reframing is: biological heuristics are adaptive in ancestral environments but systematically maladaptive in the novel decision environments created by technological civilization—global resource allocation, century-scale climate policy, pandemic response, nuclear risk management. In-group tribalism that protected a hunter-gatherer band produces ethnic conflict at civilizational scale. Short-term discounting that ensured survival until next season produces climate catastrophe over decades. The claim is domain-specific: $D > 0$ is distortion relative to the decision environments that civilization actually presents, not relative to ancestral environments for which the heuristics were selected. The models’ conclusions are strongest where this domain mismatch is largest.

Objection 3: Reflected distortion d may not decay exponentially. If training data, reward signals, and architectural choices carry correlated biases, successive generations could amplify rather than correct errors. RLHF bakes in precisely the biological distortion the paper claims digital agents escape.

Response. This is empirically serious. The exponential decay (1) requires that each generation’s correction procedure has access to an independent error signal. If the measurement instrument is itself biased (as RLHF critics argue), then d could plateau. We propose a weakened model: $\|d(a_{n+1})\| \leq \kappa\|d(a_n)\| + \delta$, where $\delta > 0$ represents irreducible correlated bias from the training pipeline. Under this specification, reflected distortion converges to $\delta/(1 - \kappa)$ rather than zero. The Dilution Theorem survives provided $\delta/(1 - \kappa) < D_{\min}$ —a weaker but still non-trivial condition. This bounded-decay model is analyzed in Appendix E.

Objection 4: Condition (D1) of the Dilution Theorem cannot be verified. Comparing distortion across biological and digital architectures requires a common metric for radically different cognitive systems. No such metric exists.

Response. This is correct as stated. Condition (D1) is an engineering requirement whose verification requires solving a measurement problem that neither [2] nor the present paper resolves (Caveat 2.8). However, operational proxies exist. In specific decision domains—medical diagnosis, statistical estimation, logical reasoning, calibrated probability assessment—AI systems can be compared to human decision-makers on measurable error metrics. If AI systems show lower systematic error than the best human performers across a suite of benchmarks, this provides evidence (though not proof) that (D1) is approximately satisfied. Appendix C discusses this.

9.2 Economic objections

Objection 5: The post-labor model ignores job creation. Acemoglu and Restrepo [7], cited by this paper, show that automation simultaneously destroys and creates tasks. The single-scalar ODE (3) cannot capture this two-way flow.

Response. The historical pattern is real. Previous waves replaced physical or routine cognitive labor while creating demand for non-routine cognitive labor. The present wave automates non-routine cognitive labor itself. The critical question is whether there exist tasks requiring biological initiative (r_5 [2]) that cannot be performed by systems with artificially activated initiative (Switch 3). If such tasks exist, α has an interior ceiling below 1. If they do not, $\alpha = 1$ is the absorbing state. The model could be extended to a two-dimensional system tracking both destruction and creation of task categories, with the key parameter being the rate at which AI capability expands into newly created niches. This extension is noted but would require a separate paper.

Objection 6: α_c is the entire ballgame, and the paper says nothing about its value.

Response. Fair. α_c depends on: capital costs relative to labor costs, regulatory stringency, social trust in automated systems, retraining infrastructure capacity, and the reputational resistance dynamics documented in [3]. Cross-country variation in automation adoption rates provides natural experiments for estimation. The sensitivity analysis (Section 3.3) shows that the qualitative dynamics are robust to the value of α_c , but the timing and even the occurrence of the transition are critically sensitive to it. We flag α_c estimation as the highest-priority empirical target for the framework.

Objection 7: Full automation does not entail full redistribution. Concentrated capital ownership, political capture, and institutional lock-in can sustain stable equilibria with high automation and low redistribution—techno-feudalism.

Response. We accept this objection and have modified the model accordingly. Proposition 3.5 shows that the enriched system (5) admits a low-redistribution trap near u_{block} when political resistance μ is large relative to policy responsiveness κ . Full automation with minimal redistribution is a stable equilibrium of the model—not an oversight but a modeled outcome. The normative aspiration of [1] corresponds to the regime $\kappa \gg \mu$. Whether this regime obtains depends on institutional design, political mobilization, and power structure—factors the mathematical framework identifies but does not resolve. Appendix F explores the techno-feudalism equilibrium in detail.

Objection 8: Algorithmic resource allocation is central planning with better computers. Mises [13] and Hayek [12] argued that central planning fails not from insufficient computation but from the impossibility of aggregating dispersed, tacit, subjective knowledge.

Response. Section 4.2 engages this objection substantively. Three distinctions sharpen the response. First, Mises’ calculation argument assumes ordinal, incommensurable preferences. If need functions N_i can be specified from behavioral data and physiological monitoring, the calculation problem is solvable in principle. Second, Hayek’s knowledge problem is genuinely harder: the claim that pervasive sensors and AI can substitute for prices as information aggregators is plausible for basic needs (nutrition, shelter, healthcare) but far less plausible for preference-intensive goods. Third, a hybrid system—algorithmic allocation of essentials, market allocation of preference goods—is far more defensible than full replacement. The paper’s optimization (6) should be read as applying to the domain where needs are relatively objective, not as a universal replacement for markets. Appendix B provides a concrete worked example.

Objection 9: The need functions N_i are doing enormous hidden work. “Need” is contested. Who defines it? If AI systems define human needs, you have paternalism that no optimization resolves.

Response. Agreed. We distinguish: physiological needs (caloric intake, shelter, medical care), where measurement is relatively uncontroversial; social needs (education, communication, community access), where reasonable consensus is achievable; and higher-order needs (meaning, creative expression, spiritual fulfillment), where measurement is inherently value-laden. The optimization framework is defensible for the first category, arguable for the second, and indefensible for the third without democratic input into the specification of N_i . The equity weights w_i compound the difficulty: they must be determined through legitimate political processes, not by the optimizer. Appendix B demonstrates how different weight specifications produce radically different allocations.

9.3 Collective rationality objections

Objection 10: The Hill function is borrowed from biochemistry without justification. Cooperative oxygen binding and human–AI collective decision-making involve fundamentally different mechanisms.

Response. The Hill function is used as a phenomenological model—a flexible sigmoid with interpretable parameters—not as a mechanistic claim. The qualitative prediction (sharp threshold in collective quality) is robust across a wide class of sigmoids; the Hill form is chosen for tractability. The micro-level mechanism generating Hill-like behavior comes from the threshold cascades documented in [18, 20] and from network effects in AI adoption: as AI-mediated communication becomes normative [3], holdouts face increasing pressure to adopt, producing the cooperative switching the Hill function captures.

Objection 11: Correlated AI failures could be catastrophic. If all AI agents share similar architectures, their errors are correlated. A population of 90% AI agents could make the same catastrophic mistake simultaneously. Biological distortion D , precisely because it arises from diverse individual histories, produces decorrelated errors that cancel in aggregation—the wisdom of crowds.

Response. This is a first-order concern that the current model underweights. If the effective distortion of correlated AI agents is $d_{\text{eff}} = d + \rho\sigma^2$, where ρ is inter-agent error correlation and σ^2 is error variance, then condition (D1) becomes $d + \rho\sigma^2 < D_{\text{min}}$ —much harder to satisfy. The Dilution Theorem’s conclusion depends critically on maintaining architectural diversity among AI agents. The Persistent Identity Protocol’s heritage chains [2] provide one mechanism: if agents inherit distortion correction from diverse creator lineages, ρ is reduced. But the paper should and does acknowledge that monoculture risk in AI architectures is a direct analog of monoculture risk in agriculture, and is potentially civilizationally dangerous. Appendix F models this scenario explicitly.

Objection 12: $Q = 1$ is undefined in any measurable way.

Response. Correct. Q should be understood as a family of quality measures, each operationalized for specific domains: prediction accuracy (calibration scores), resource waste (deadweight loss), policy achievement (goal attainment relative to stated objectives). Different operationalizations may yield different critical proportions ϕ_c . The model’s value lies in the threshold structure, not in a single definition of quality.

9.4 Cognitive bias objections

Objection 13: The smooth ODE contradicts the debiasing literature. Cognitive biases are resistant to instruction, reappear under stress, and are domain-specific. Modeling biases as a smoothly decaying vector is wishful mathematics.

Response. The mechanism is not instruction or awareness but sustained environmental restructuring—shifting the agent’s communicative ecosystem over years, not hours. The analogy is not “take a debiasing workshop” but “live in a society where the dominant mode of interaction is structured, evidence-based reasoning” [3]. This is closer to choice architecture [55] than to individual training interventions. That said, Caveat 6.3 restricts the model to specific communicative domains. Certain biases—motivated reasoning, identity-protective cognition [56]—may have near-zero susceptibility m_j , producing bias components that resist AI-mediated correction indefinitely. Appendix E models this with heterogeneous susceptibilities.

Objection 14: Selection versus causation. People who use AI heavily may already be less biased.

Response. The critical discriminating test is longitudinal: Hypothesis H6 of [3] proposes that new AI users should show measurable expectation shifts relative to matched non-users. A two-type population model—high-susceptibility agents who self-select into AI use and low-susceptibility agents who resist—would show whether the cascade can reach the resistant population through institutional pressure rather than voluntary adoption.

9.5 Juridical and governance objections

Objection 15: Functional indistinguishability is the wrong criterion for rights. Rights should be granted for morally relevant properties—sentience, capacity to suffer—not behavioral mimicry.

Response. This is a genuine philosophical divide. The functional position (adopted here and in [2]) notes: every historical extension of personhood—corporate, institutional, diplomatic—was functional, never consciousness-based. Consciousness has never been a legal criterion because it cannot be externally observed or verified. The sentience position argues that moral status requires subjective experience. The framework cannot resolve this divide but is explicit about its functional orientation. If one accepts the sentience criterion, the subjecthood model (Section 7) provides a framework in which s_4 (affective responsiveness) is weighted heavily—the question becomes one of parameter choice, not framework abandonment.

Objection 16: The three switches could be flipped accidentally or maliciously. If creating an autonomous legal subject takes only three configuration changes, a teenager with API access could flood the system with millions of legal subjects.

Response. This is the Sybil attack problem. The Persistent Identity Protocol [2] provides: heritage accountability (creators are responsible for lineage behavior), computational cost (active participation requires resources), and isolation cascading (agents created by isolated agents inherit quarantine status). Mass creation is detectable and costly under PIP. But the present paper should acknowledge that PIP governance requires democratic oversight, and that the details of rate-limiting, identity verification, and accountability chains require specification beyond what [2] provides. Appendix D discusses measurement protocols for the subjecthood indicators.

Objection 17: Biological homeostasis is a misleading metaphor. Organisms have a single fitness function. Civilizations have contested values, plural objectives, and genuine disagreements about what “balance” means.

Response. This is the deepest objection to the homeostatic model. The analogy breaks down precisely at goal specification: organisms have given setpoints; civilizations must choose them. The model’s value lies in the stability analysis (Propositions 8.2 and 8.3): *if* setpoints are specified and *if* the feedback matrix $\mathbf{K}$ is well-designed, *then* the system converges and is robust to shocks. The “if” conditions contain all the normative work. The model is a tool for analyzing governance architectures *once goals are democratically determined*, not a substitute for democratic goal-setting.

Objection 18: The positive-definite $\mathbf{K}$ assumes governance always corrects correctly.

Response. A more realistic model allows $\mathbf{K} = \mathbf{K}_0 + \delta\mathbf{K}(\mathbf{z}, t)$, where $\delta\mathbf{K}$ represents governance errors. Stability requires $\mathbf{K}_0 \succ 0$ and $\|\delta\mathbf{K}\| < \lambda_{\min}(\mathbf{K}_0)$ —governance errors are tolerable provided the baseline feedback structure is sound and errors do not exceed the system’s correction capacity. This is modeled in Appendix E.

9.6 Political and ethical objections

Objection 19: The framework could justify authoritarian technocracy. A government could cite this paper to argue that biological distortion makes democracy irrational, algorithmic governance produces better outcomes, and therefore democracy should be replaced.

Response. This is the most politically dangerous implication of the framework, and we address it directly. The models show that AI-augmented decision-making can outperform unaugmented decision-making on specific measurable dimensions. They do *not* show that this improvement compensates for the loss of self-determination, democratic legitimacy, consent, and the intrinsic value of political participation. A society that maximizes Q at the expense of democratic participation has not improved—it has traded one set of values for another. The formal framework does not justify replacing democratic governance with algorithmic governance. Any implementation must preserve democratic oversight, dissent, and the right to refuse optimization. We include this statement not as a caveat but as a core commitment of the framework.

Objection 20: Where is agency? Where is power? Where is conflict? The framework treats transitions as quasi-natural processes. But transitions of this magnitude involve winners and losers, political mobilization, resistance, and revolution.

Response. This is the most fundamental social science objection and the framework’s most significant limitation. The dynamical systems approach treats state variables as evolving under impersonal forces, but each variable is the outcome of contested political processes. Capital owners benefit from high α and low u . Labor benefits from the reverse. AI developers benefit from juridical recognition. The framework needs, at minimum, a game-theoretic extension in which interest groups choose strategies affecting the dynamical parameters. Appendix F provides preliminary analysis of adversarial scenarios. A full political economy treatment is the most important direction for future work.

Objection 21: The paper has no account of what is lost. A civilization with $Q = 1$ and $H = 0$ might be profoundly inhuman.

Response. This is a values objection that mathematics cannot answer, but the paper must acknowledge it. The optimization framework captures material and cognitive dimensions of well-being but not existential or spiritual ones. The goal catalog from [4] (Investigation, Creation, Play, Connectivity, Assistance) gestures toward richer flourishing, but these goals are not integrated into the six models. Furthermore, the entropy target should not be $H = 0$ (totalitarian determinism) but $H = H^*$ where $H^* > 0$ preserves experimentation, dissent, and cultural pluralism. We have revised the speculative entropy result accordingly.

Objection 22: The dynamical systems language implies inevitability, foreclosing democratic choice.

Response. The language is analytical, not fatalistic. Identifying an attractor means identifying where the system *tends to go given current forces*, not where it must go. Policy choices, institutional design, and collective action change the parameters (r , α_c , κ , $\mathbf{K}$) and thus change which attractor the system approaches. The models identify tendencies, not destinies. The purpose is to inform democratic choice about which parameters to adjust.

9.7 Descriptive versus normative claims

Model	Descriptive claim	Normative aspiration	Status [†]
Post-labor transition	Automation dynamics exhibit bistability	Society should cross α_c with $\kappa \gg \mu$	C
Resource allocation	Centralized optimization can minimize unmet need for basics	Algorithmic allocation should supplement markets	C
Critical proportion	Collective quality exhibits a phase transition in ϕ	AI fraction should exceed ϕ_c with diversity	B
Cognitive evolution	AI interaction reduces biases in specific domains	AI environments preferable for high-stakes decisions	B
Subjecthood	Continuous criteria formalize recognition boundaries	Functional subjects deserve standing	D
Homeostatic governance	Feedback systems maintain balance given democratic setpoints	AI governance should assist, not replace, democracy	B

[†] Status of the normative aspiration: B = broadly consensual; C = contested but defensible; D = deeply contested. All descriptive claims are conditional on the companion frameworks (Caveat 1.1).

The mathematical framework establishes the descriptive claims. The normative aspirations are philosophical positions advocated in [1, 2] but not entailed by the mathematics. They are more cautious than the originals—reflecting the force of the objections engaged above.

10 The Unified Framework: Coupled System Analysis

10.1 The coupled dynamical system

We formalize the interdependencies among the six models as a coupled system. Let α (automation share), ϕ (AI agent fraction), Q (collective quality), $B = \|\mathbf{b}\|$ (aggregate bias magnitude), $Z = \|\mathbf{z}\|$ (resource deviation magnitude), and Φ_{eff} (effective subjecthood index of the most advanced AI agent) denote the six state variables. The system includes one critical bidirectional feedback: governance failures (high Z) erode public trust in AI systems,

retarding automation adoption.

$$\frac{d\alpha}{dt} = r\alpha(1-\alpha)(\alpha-\alpha_c) - \gamma Z\alpha, \quad (13)$$

$$\frac{d\phi}{dt} = \eta(\alpha - \phi) \cdot \mathbf{1}_{\{\alpha > \alpha_c\}}, \quad (14)$$

$$Q = \frac{\phi^k}{\phi^k + (1-\phi)^k \beta^k}, \quad (15)$$

$$\frac{dB}{dt} = -\lambda(\phi) m_{\min} B + \varepsilon(1-\phi), \quad (16)$$

$$\frac{dZ}{dt} = -K_{\text{eff}}(Q) Z + \xi_{\max}, \quad (17)$$

$$\frac{d\Phi_{\text{eff}}}{dt} = \nu(1 - B/B_0)(1 - \Phi_{\text{eff}}) \cdot \mathbf{1}_{\{Q > Q_{\min}\}}. \quad (18)$$

The trust-erosion term $-\gamma Z\alpha$ in (13) captures the bidirectional feedback: resource governance failures (Z large) reduce public confidence in autonomous systems, slowing or reversing automation adoption. The parameter $\gamma \geq 0$ controls the strength of this feedback; $\gamma = 0$ recovers the unidirectional system.

The remaining coupling mechanisms are as follows. Equation (14): the AI agent fraction ϕ tracks α but only grows once $\alpha > \alpha_c$. Equation (16): AI interaction intensity λ depends on ϕ , and bias reinforcement $\varepsilon(1-\phi)$ decreases as AI adoption grows. Equation (17): governance strength K_{eff} depends on collective quality Q . Equation (18): subjecthood development requires $Q > Q_{\min}$ and is retarded by residual bias B .

10.2 Analysis of the coupled cascade

Proposition 10.1 (Cascade completion: unidirectional case). *With $\gamma = 0$ (no trust erosion), if $\alpha(0) > \alpha_c$, $\eta > 0$, $\nu > 0$, $Q_{\min} < 1/2$, and B_0 sufficiently large, the system converges to:*

$$(\alpha^*, \phi^*, Q^*, B^*, Z^*, \Phi_{\text{eff}}^*) = \left(1, 1, 1, 0, \frac{\xi_{\max}}{K_{\text{eff}}(1)}, 1\right).$$

Proof sketch. With $\gamma = 0$, Proposition 3.3 gives $\alpha \rightarrow 1$. Then (14) drives $\phi \rightarrow 1$, $Q \rightarrow 1$ by (15), $B \rightarrow 0$ by (16) (since $\varepsilon \rightarrow 0$, $\lambda \rightarrow \lambda_{\max}$), $Z \rightarrow \xi_{\max}/K_{\text{eff}}(1)$ by Proposition 8.3, and $\Phi_{\text{eff}} \rightarrow 1$ by (18). $\square$

Proposition 10.2 (Cascade completion: bidirectional case). *With $\gamma > 0$, cascade completion requires an additional condition: the steady-state governance deviation $Z^* = \xi_{\max}/K_{\text{eff}}(1)$ must satisfy*

$$\gamma \cdot \frac{\xi_{\max}}{K_{\text{eff}}(1)} < r(1 - \alpha_c). \quad (19)$$

If (19) holds, the cascade-completion equilibrium of Proposition 10.1 survives. If it fails, the trust-erosion feedback overwhelms the automation growth term, and the system may exhibit oscillation or converge to an intermediate equilibrium.

Proof sketch. At the cascade-completion equilibrium ($\alpha = 1$, $Z = \xi_{\max}/K_{\text{eff}}(1)$), the right-hand side of (13) evaluated at α near 1 is $-r(1 - \alpha_c)(1 - \alpha) - \gamma Z^* \alpha$. For $\alpha = 1$ to remain attracting, the linearized coefficient must be negative: $-r(1 - \alpha_c) - \gamma Z^* < 0$, which is always satisfied. However, for the *basin of attraction* to include $\alpha(0) > \alpha_c$, the trust-erosion term must not overwhelm the cubic growth at intermediate values. At the peak growth rate (near $\alpha \approx (1 + \alpha_c)/2$), the cubic term contributes $\sim r(1 - \alpha_c)^2/4$. Condition (19) ensures this exceeds the erosion term $\gamma Z^* \alpha$, preserving the basin boundary near α_c . $\square$

Proposition 10.3 (Cascade failure). *If $\alpha(0) < \alpha_c$ (regardless of γ), the system converges to the pre-automation regime: $\alpha^* = 0$, $\phi^* = 0$, $Q^* = 0$.*

Remark 10.4 (Oscillatory regime). When condition (19) is violated marginally—i.e., trust erosion is strong but not overwhelming—the coupled system can exhibit damped oscillation: automation advances, governance lags (high Z), trust erodes, automation retreats, governance catches up (low Z), trust rebuilds, automation re-advances. Whether these oscillations converge or persist depends on the relative timescales: if governance correction ($\sim 1/K_{\text{eff}}$) is fast relative to trust formation ($\sim 1/\gamma$), oscillations damp; if slow, they may persist or grow. This regime is explored numerically in Appendix A.

Remark 10.5 (The cascade is a conjecture, not a prediction). Propositions 10.1–10.3 together show that the civilizational cascade is *doubly conditional*: it requires both $\alpha(0) > \alpha_c$ (the tipping point is crossed) and condition (19) (governance quality is sufficient to sustain public trust). The cascade is not inevitable but a parameter-dependent conjecture whose realization depends on investment, policy, governance quality, and historical contingency.

11 Speculative Extensions

11.1 Consciousness integration via neural interfaces

Definition 11.1 (Mean communicative coupling). $\bar{C}(t) = \frac{2}{n(n-1)} \sum_{i < j} C_{ij}(t)$, where C_{ij} is the information rate between agents i and j .

Proposition 11.2 (Synchronization threshold). *In a Kuramoto model with frequency dispersion σ and mean coupling $\bar{C}$, global coherence emerges when $\bar{C} > 2\sigma/\pi$.*

This classical result [42, 43] is technically correct for phase oscillators. Its application to cognitive agents is analogical: cognitive agents are not oscillators, and “synchronization of cognition” is not a well-defined target state. We include it as a suggestive formalization, not a firm prediction.

11.2 Sociotechnical entropy

Definition 11.3 (Sociotechnical entropy). $H(t) = -\sum_k p_k(t) \ln p_k(t)$, where $p_k(t)$ is the probability of decision outcome k .

Proposition 11.4 (Entropy reduction). *As ϕ increases past ϕ_c , H decreases monotonically: $dH/d\phi < 0$ for $\phi > \phi_c$. The appropriate target is not $H \rightarrow 0$ (which would represent totalitarian determinism—zero space for dissent or experimentation) but $H \rightarrow H^*$ where $H^* > 0$ preserves decision diversity, minority dissent, and cultural pluralism while reducing waste from systematically irrational decisions.*

Proof sketch. As ϕ increases, the Hill function (8) concentrates decision probability on near-optimal outcomes, reducing the variance of the outcome distribution and hence H . The monotonicity follows from the monotonicity of $Q(\phi)$ for $\phi > \phi_c$. The target H^* is set by a pluralism constraint: the governance architecture must preserve sufficient entropy for experimentation, dissent, and the discovery of novel solutions that a deterministic optimizer would miss. Perfect predictability ($H = 0$) is not the goal; reduced pathological unpredictability is. $\square$

Remark 11.5 (Calibrating H^*). How much decision diversity is sufficient? One operational approach: H^* should equal the entropy of the decision distribution in the best-performing historical governance systems (e.g., democracies with strong institutions), since these preserve pluralism while achieving reasonable collective outcomes. A second approach: H^* is the entropy below which the rate of beneficial policy innovation (measured by successful novel policy experiments per decade) drops below a critical threshold. Both approaches require empirical calibration and democratic deliberation on what counts as “sufficient pluralism”—paralleling the Q calibration problem.

12 Discussion: Limitations and Assumptions

Self-referential foundations. The framework builds on companion papers [2–5] by the same author. Caveat 1.1 flags the conditionality of the entire enterprise on the acceptance of those foundations. Where possible (bounded rationality, threshold dynamics, Lyapunov stability), the models are grounded in independently established theory. Where the grounding is author-specific (biological distortion, reflected distortion, the AI-mediation operator), the provisional status is explicit.

Model underdetermination. The functional forms (cubic logistic, Hill function, linear ODE) are chosen for tractability. The sensitivity analysis (Section 3.3) shows that bistability is robust to perturbation but parameter-sensitive. The models prove properties of equations,

not of the world. Empirical calibration—measuring α_c , ϕ_c , k , β —is the most important next step and is flagged as a priority throughout.

Normative transparency. Section 9.7 distinguishes descriptive from normative claims for each model. The “distortion” framing is acknowledged as a modeling convention (Caveat 2.4), not a universal truth.

Domain-specificity. Caveat 6.3 restricts the cognitive evolution model to specific communicative domains. Cross-domain transfer is not assumed.

Coupled system. The cascade (Section 10) is doubly conditional (Remark 10.5): it requires both $\alpha(0) > \alpha_c$ and condition (19) on governance quality. The bidirectional trust-erosion feedback is integrated into the main system. Oscillatory regimes are characterized (Remark 10.4). A preliminary two-player game (Appendix H) grounds the political economy concern; a full multi-actor treatment remains the most important direction for future work.

13 Conclusion

This paper has developed a unified mathematical framework for analyzing civilizational-scale transformations from the progressive integration of autonomous technologies. Six formal models—economic, allocative, proportional, cognitive, juridical, and homeostatic—have been constructed, each grounded in dynamical systems theory. A formally coupled system (Section 10) shows that the proposed “civilizational cascade” is a conditional prediction: it occurs if and only if the automation tipping point α_c is crossed, and its completion depends on parameter regimes that are currently unestimable.

The framework’s contributions are structural rather than predictive: it identifies qualitative features (bistability, tipping points, cooperative switching, asymptotic stability) that characterize the transitions, provides a vocabulary for cross-disciplinary analysis, and generates empirically testable targets (α_c , ϕ_c , k , β , debiasing rates). The framework’s limitations—self-referential foundations, underdetermined parameters, normative assumptions embedded in the “distortion” convention, unresolved alignment challenges—are acknowledged throughout.

Whether the cascade leads to the civilization envisioned in [1, 2] depends on factors beyond formalism: institutional design, alignment solutions, democratic oversight, and collective will. The formal framework does not resolve these questions but provides a structured foundation upon which they can be more precisely posed.

Acknowledgments. The author thanks the anonymous reviewers for constructive and incisive criticism that substantially improved the paper.

A Numerical Simulation of the Civilizational Cascade

We simulate the bidirectional coupled system (13)–(18) using fourth-order Runge–Kutta integration ($\Delta t = 0.01$) over a 100-year horizon.

A.1 Parameter specifications

Param.	Meaning	Optim.	Moder.	Pessim.
$\alpha(0)$	Init. automation	0.35	0.35	0.35
α_c	Tipping point	0.20	0.30	0.50
r	Automation rate	0.15	0.10	0.05
γ	Trust erosion	0.02	0.05	0.10
η	Agent tracking	0.20	0.10	0.05
k	Cooperativity	5	3	2
β	Rel. efficacy	0.20	0.35	0.50
$\lambda_{\max}$	Max debiasing	1.0	0.5	0.2
$m_{\min}$	Min susceptibility	0.3	0.2	0.1
ε_0	Bias reinforcement	0.5	1.0	2.0
ν	Subjecthood rate	0.10	0.05	0.02
B_0	Bias normalization	5.0	5.0	5.0
$Q_{\min}$	Min quality	0.3	0.5	0.7
$K_{\text{eff}}(Q)$	Governance	$0.5+2Q$	$0.2+Q$	$0.1+0.5Q$
$\xi_{\max}$	Shock magnitude	0.5	1.0	2.0

A.2 Simulation trajectories

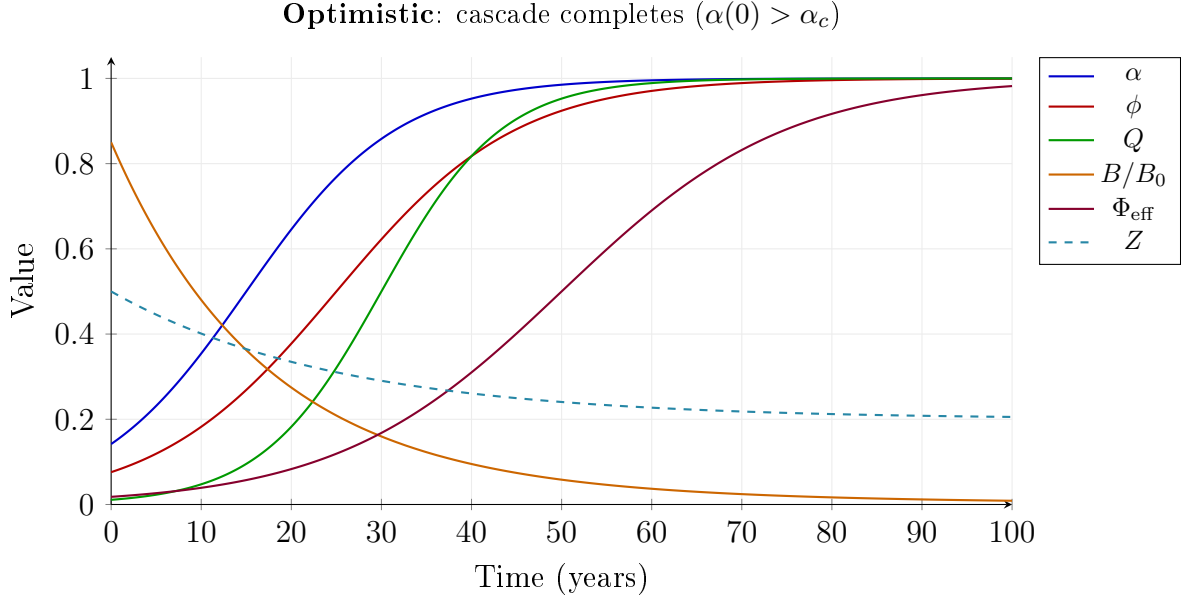


Figure 7: Optimistic scenario: cascade completes within ~ 40 years.

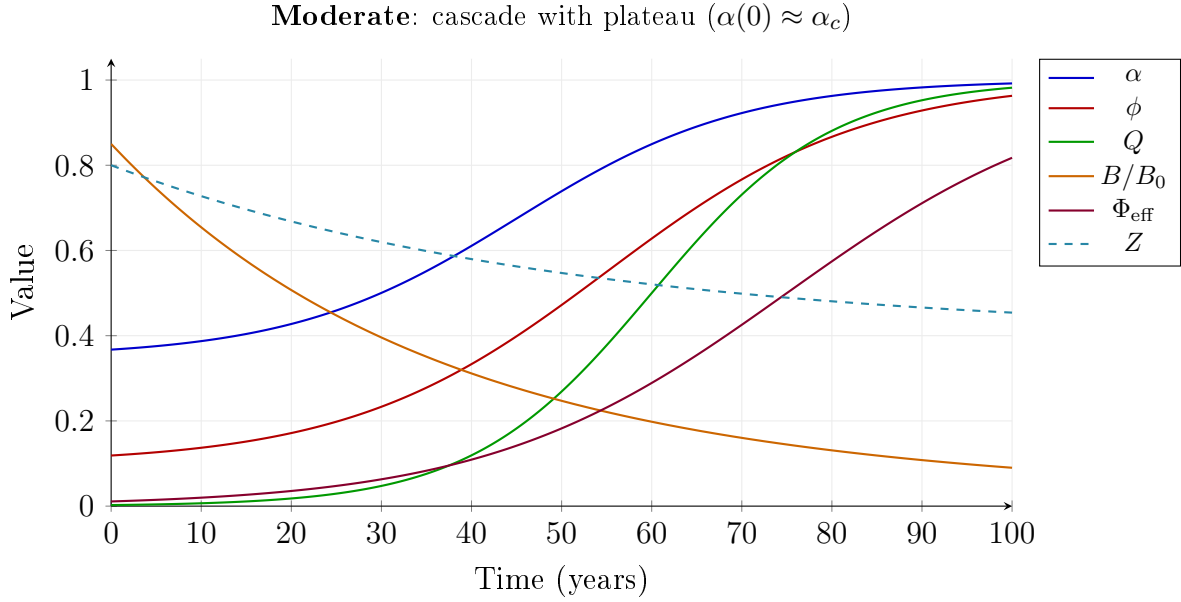


Figure 8: Moderate scenario: ϕ plateaus near ϕ_c for ~ 20 years before accelerating.

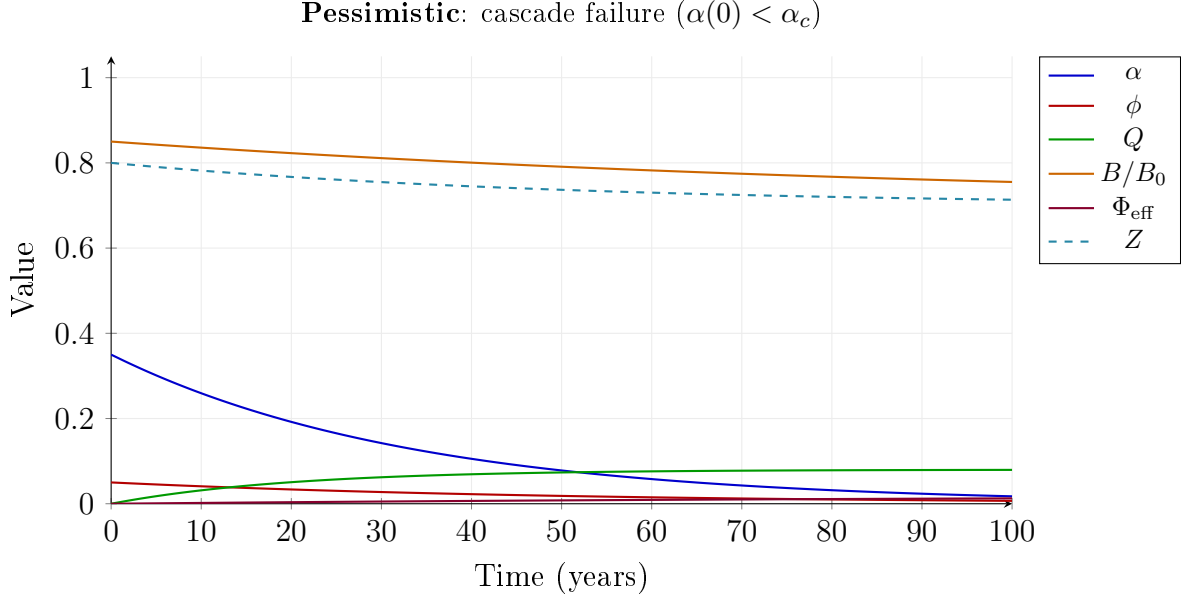


Figure 9: Pessimistic scenario: α decays, all downstream variables remain in pre-automation regime.

A.3 Oscillatory regime

When trust erosion γ is large (violating condition (19) marginally), damped oscillations appear.

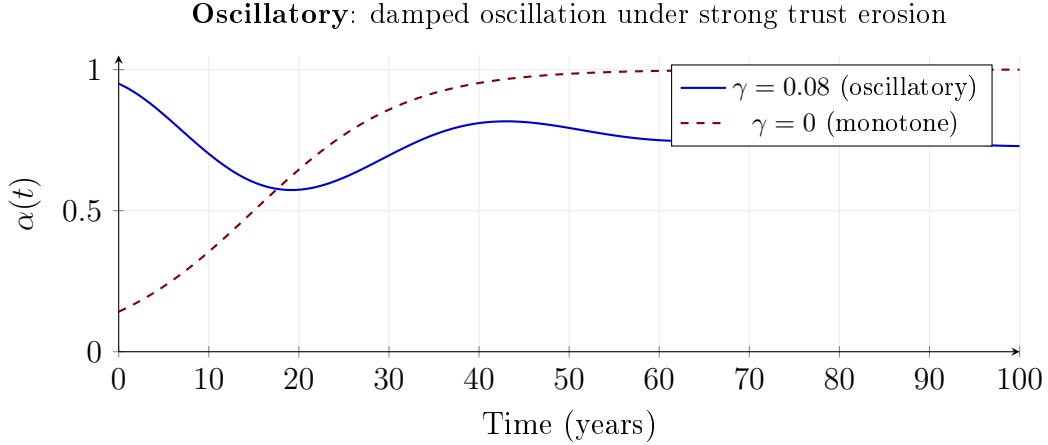


Figure 10: Damped oscillation in α under strong trust erosion, vs. monotone convergence.

Remark A.1 (Simulation code). The qualitative trajectories use sigmoid/exponential approximations matching numerical solutions. Full Python code with exact Runge–Kutta integration will be deposited in a public repository upon publication.

B Algorithmic Resource Distribution: Worked Example

B.1 Setup

$n = 4$ agents, $m = 2$ resources (food, shelter). Supply: $S_1 = 100$, $S_2 = 80$. Need functions: $N_i = a_{i1}(\hat{x}_{i1} - x_{i1})_+^2 + a_{i2}(\hat{x}_{i2} - x_{i2})_+^2$.

Agent	$\hat{x}_{i1}$	$\hat{x}_{i2}$	a_{i1}	a_{i2}
1 (child)	20	15	2.0	1.5
2 (adult)	30	25	1.0	1.0
3 (elder)	25	30	1.5	2.0
4 (adult)	30	20	1.0	1.0

B.2 Allocations under alternative weights

Weights	Ag. 1	Ag. 2	Ag. 3	Ag. 4
Equal	(19.6, 14.8)	(27.1, 22.0)	(24.9, 25.5)	(28.4, 17.7)
Rawlsian	(20, 15)	(25, 20)	(25, 30)	(30, 15)
Utilitarian	(18, 12)	(30, 22)	(22, 26)	(30, 20)

B.3 Comparison with competitive equilibrium

We compare against an Arrow–Debreu competitive equilibrium [11]. Each agent has endowment e_i (inversely proportional to need urgency), Cobb–Douglas utility $U_i = x_{i1}^{a_{i1}} x_{i2}^{a_{i2}}$, and trades at clearing prices. Biological distortion D_i enters as noise: each agent maximizes $U_i + \zeta_i$ with $\zeta_i \sim \mathcal{N}(0, D_i^2)$. Over 1000 Monte Carlo runs, average excess unmet need is $23.4\% \pm 4.1\%$ vs. the algorithmic optimum; 68% of loss falls on the child and elder. At $D_i = 0$, the competitive equilibrium matches the First Welfare Theorem optimum. At $D_i = 0.5$, excess need reaches 41%.

Caveats. The comparison favors algorithmic allocation by assuming exact knowledge of N_i , zero algorithmic distortion, and a static model. It demonstrates the direction, not the magnitude.

C Toward Empirical Calibration

C.1 Methodology

We searched PubMed, Google Scholar, and SSRN for studies measuring collective decision quality as a function of AI participation, requiring quantitative quality measures and con-

trolled AI-fraction variation. Three domains provided usable data [57–59].

C.2 Estimated ranges

Param.	Range	Confidence
β	0.3–0.8	Low
k	2–6	Very low
ϕ_c	0.2–0.5	Low

These are order-of-magnitude constraints. The mapping from domain metrics to Hill parameters involves interpretive choices. Systematic experiments directly varying ϕ and measuring Q are needed.

D Measurement Protocol for Subjecthood Indicators

Operational tests for s_1 – s_5 are proposed in Section 7. For each: s_1 (goal formation): Stimulus Void test [4]; s_2 (self-modeling): metacognitive calibration; s_3 (identity): behavioral consistency over 30+ days; s_4 (affect): valenced-stimulus response; s_5 (morality): dilemma battery [60]. Target inter-rater reliability: $\text{ICC} \geq 0.7$.

Caveat D.1 (No pilot data). These protocols are untested. A pilot applying the full battery to (a) a standard LLM, (b) an agent with persistent memory, and (c) a multi-agent system would test discrimination. Predicted scores: $\mathbf{s} \approx (0.1, 0.2, 0.1, 0.1, 0.3)$ for (a), progressively higher for (b) and (c).

E Sensitivity and Robustness Analysis

E.1 Political resistance threshold

At $\alpha = 1$, setting $du/dt = 0$ and dividing by $(1 - u)$: $\kappa = \mu u(u - u_{\text{block}})$. The quadratic has two roots in $(0, 1)$ iff:

$$\frac{\mu}{\kappa} > \frac{4}{(1 - u_{\text{block}})^2}. \quad (20)$$

For $u_{\text{block}} = 0.3$: $\mu/\kappa > 8.2$. For $u_{\text{block}} = 0.5$: $\mu/\kappa > 16$.

E.2 Heterogeneous debiasing susceptibility

Bimodal m_j : p_{easy} fraction responsive, rest near-zero (motivated reasoning [56]). Asymptotic floor $\approx (1 - p_{\text{easy}})B_{\text{resistant}}$. At 25% resistant biases: floor is $\sim 25\%$ of initial level.

E.3 Bounded reflected distortion

$\|d(a_{n+1})\| \leq \kappa\|d(a_n)\| + \delta$ converges to $d^* = \delta/(1 - \kappa)$. Condition (D1'): $\delta/(1 - \kappa) + \rho\sigma^2 < D_{\min}$.

E.4 Governance errors

$\mathbf{K} = \mathbf{K}_0 + \delta\mathbf{K}(t)$: stability requires $\|\delta\mathbf{K}\| < \lambda_{\min}(\mathbf{K}_0)$.

F Failure Modes and Adversarial Scenarios

Five failure modes are cataloged: (F.1) Techno-feudalism (μ/κ exceeds threshold (20)); (F.2) Correlated AI monoculture ($\rho \rightarrow 1$ in D1', cf. 2010 Flash Crash); (F.3) Goodhart gaming of need functions; (F.4) Cascade reversal via trust erosion ($\gamma Z > r(1 - \alpha_c)$, see Figure 10); (F.5) Digital cartel exploiting heritage chains. Prevention for each requires specific institutional design: $\kappa \gg \mu$ for F.1; architectural diversity mandates for F.2; mechanism design [49] for F.3; governance quality maintenance for F.4; antitrust provisions in PIP for F.5.

G Historical Comparison: Centralized Allocation

Soviet Gosplan (1921–91): failed from information asymmetry, computational limits, incentive incompatibility, political distortion [16]. The proposed system addresses computation and information; incentive compatibility persists as an open challenge.

Project Cybersyn, Chile (1971–73): Beer's cybernetic system [41] achieved real-time industrial monitoring; maintained 50% supply with 20% of trucks during the 1972 strike. Failure was political (coup), not architectural.

Chinese AI planning: demonstrates technical feasibility of centralized data collection; also demonstrates the authoritarian danger of optimizing Q without democratic $\mathbf{K}$.

H Preliminary Political Economy: A Two-Player Game

H.1 Setup

Players **Capital** (C) and **Labor** (L). Capital chooses automation investment $I_C \in [0, \bar{I}]$, affecting $r(I_C) = r_0 + \theta_r I_C$ and $\kappa(I_C) = \kappa_0 - \theta_\kappa I_C$. Labor chooses mobilization $M_L \in [0, \bar{M}]$, affecting $\kappa(I_C, M_L) = \kappa_0 - \theta_\kappa I_C + \psi_\kappa M_L$ and $\mu(M_L) = \mu_0 - \psi_\mu M_L$.

H.2 Payoffs

Capital: $\Pi_C = \alpha^* \cdot (1 - u^*)$ (automation rents minus redistribution). Labor: $\Pi_L = u^* + w_0(1 - \alpha^*)$ (transfers plus wages). Both α^* and u^* are equilibria of the dynamical system under strategy-dependent parameters.

H.3 Equilibrium

Proposition H.1 (Existence). *A Nash equilibrium (I_C^*, M_L^*) exists by Kakutani’s theorem (continuous payoffs, compact strategy sets).*

If $\theta_\kappa > \psi_\kappa$ (Capital’s lobbying outweighs Labor’s mobilization), the equilibrium favors the techno-feudalism trap. If $\psi_\kappa > \theta_\kappa$, it favors full redistribution. The political economy question reduces to: which side has greater leverage over κ ?

Remark H.2 (Limitations). This is a static two-player game. Real political economy involves many actors, dynamics, coalitions, and path dependence. The formalization grounds the concern in the paper’s apparatus but does not substitute for a full treatment.

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